

## Problem 6

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**Problem 1** A piecewise defined function  $f$  is given by

$$f(x) = \begin{cases} \frac{2x-3}{x-2} & \text{if } x < 2 \\ \frac{x^2-5x+6}{x^2-4} & \text{if } x > 2 \end{cases}$$

Determine the form of the limit, then find the limit.

(a)  $\lim_{x \rightarrow 2^+} f(x)$

**Explanation.** Since  $\lim_{x \rightarrow 2^+} f(x) = \lim_{x \rightarrow 2^+} \frac{x^2-5x+6}{x^2-4}$ , the form of the limit is  $\frac{0}{0}$ .

$$\lim_{x \rightarrow 2^+} f(x) = \lim_{x \rightarrow 2^+} \frac{x^2-5x+6}{x^2-4} = \lim_{x \rightarrow 2^+} \frac{(x-3)(x-2)}{(x-2)(x+2)} = \lim_{x \rightarrow 2^+} \frac{x-3}{x+2} = -\frac{1}{4}$$

(b)  $\lim_{x \rightarrow 2^-} f(x)$

**Explanation.** Since  $\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^-} \frac{2x-3}{x-2}$ , the form of the limit is  $\frac{\#}{0}$ .

$$\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^-} \frac{2x-3}{x-2} = -\infty,$$

since the numerator positive, the denominator negative and goes to 0.

(c)  $\lim_{x \rightarrow 2} f(x)$

**Explanation.** The limit does not exist, because  $\lim_{x \rightarrow 2^+} f(x) \neq \lim_{x \rightarrow 2^-} f(x)$ .